Corridor MPC for Multi-Agent Inspection of Orbiting Structures

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***Abstract*— In this work, we propose an extension of the previously introduced Corridor Model Predictive Control scheme for high-order and distributed systems, with an application for on-orbit inspection. To this end, we leverage high order control barrier function (HOCBF) constraints as a suitable control approach to maintain each agent in the formation within a *safe* corridor from its reference trajectory. Recursive feasibility of the designed MPC scheme is tested numerically, while suitable modifications of the classical HOCBF constraints definition are introduced such that safety is guaranteed both in sampled and continuous time. The designed controller is validated through computer simulation in a realistic inspection scenario of the International Space Station.**

I. INTRODUCTION

The application of multi-agent systems (MAS) design to solve complex robotics tasks has received increasing attention in the past few decades [1], [2]. Examples of successful MAS control paradigms for terrestrial and aerial applications are extensive in the literature due to the broad application range. The advantages of MAS design include redundancy and robustness to single agent failure, reduced complexity in single agent hardware and the possibility to accomplish complex interactions among heterogeneous agents. These same advantages are of critical importance for the next generation of planetary exploration, on-orbit servicing and construction mission concepts, to mention a few [3]–[5].

In this work, we propose a solution to the problem of multi-agent inspection of on-orbit space vehicles using unmanned autonomous spacecrafts [6]. The design of such safe and reliable orbit-servicing missions concepts has recently revealed increasing attention. In particular, the ability to autonomously inspect, repair or de-orbit space vehicles has the potential to lower the cost of replacing commercial space assets while reducing the possibility of increasing the population of space debris orbiting the Earth. This motivates the efforts towards a fully autonomous inspection of such space assets. In this paper, the inspection task is structured as follows: it is assumed that a formation of CubeSats, called the *inspectors* [6], is deployed into a set of Passive Relative Orbits (PRO) [7] around a space vehicle orbiting a planetary body in a nearly circular orbit. Each inspector is independently controlled through a sampled-data model predictive controller (MPC), which is applied to track the assigned PRO under the influence of orbital perturbations. Appropriate High Order Control Barrier Function (HOCBF) constraints are introduced inside the MPC control scheme to constrain the system inside worst-case velocity and position tracking error bounds for each inspector in the formation.

The application of sampled-data CBF inside a Finite Horizon Optimal Control scheme (FHOC) such as MPC has already been explored in [8]. Still, safety in between discrete time steps is not analyzed. This problem is first addressed in [9], [10] instead, where suitable corrective terms are added to the continuous time CBF constraint formulation to ensure safety in between time steps. However, only first relative degree systems under zero disturbances are analyzed in [10], while only time-invariant dynamics are analyzed in [9]. In [11], the results from [9], [10] are unified under a unique framework and expanded to first order relative degree systems with time-varying dynamics and subject to state disturbances.

Based on the so called Corridor MPC (CMPC) scheme developed by [11], the contributions of this work are as follows: i) expand the definition of sampled-data CBF in [11] to the case of higher relative degree systems, ii) apply the expanded CMPC control scheme to a realistic multi-agent inspection mission of a space vehicle (SV). The recursive feasibility of the derived CMPC control scheme is then shown numerically following an approach similar to [12]. We remark that only the planning and control parts of the inspection mission are analyzed here, while problems inherent to the visual inspection of the space vehicle are topics of future work.

The manuscript is divided as follows: In Section II, we review the fundamentals of relative spacecraft dynamics and HOCBF for safety-critical dynamical systems. In Section III, we formally present the inspection problem and in Section IV we propose the new definition of sample data HOCBF as an expansion to the work in [9]–[11]. In Section V, we present the full Corridor MPC for high-order systems. Lastly, in Sections VI-VII we show a numerical simulation proving the applicability of the proposed solution in a realistic inspection of the International Space Station (ISS), followed by concluding remarks and possible extensions of this work.

Notation: Small, bold letters represent vectors. Matrices are denoted by bold, capital letters. Regular letters denote scalars. The symbol $\wedge$ is the logical *and* operator while $\times$ indicates the cross product among two vectors of the same dimension or the set product among sets. Calligraphic letters denote reference frames, and the basis vectors of a frame $\mathcal{A}$ are denoted $\{\mathbf{a}_x, \mathbf{a}_y, \mathbf{a}_z\}$. The weighted vector norm $\sqrt{\mathbf{x}^T\mathbf{A}\mathbf{x}}$ is denoted $\|\mathbf{x}\|_{\mathbf{A}}$. The notation $\|\cdot\|$ represents

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the standard Euclidean 2-norm, while the hat symbol $\hat{(\cdot)}$ on top of a vector quantity denotes a unitary (unit-norm) vector. A continuous function, $\alpha(\cdot) : \mathbb{R} \to \mathbb{R}$ is an extended class $\mathcal{K}$-function if it is strictly increasing and $\alpha(0) = 0$, while $\alpha(\cdot)$ is of class C^l if it is l times continuously differentiable in its argument. Given a sampling time $\Delta t \geq 0$, we define a discrete time instant as $k\Delta t \triangleq t_0 + k\Delta t$ with $k \in \mathbb{N}_0 = \mathbb{N} \cup \{0\}$ and initial reference time $t_0 \in \mathbb{R}_{\geq 0}$. We will use the notation $\boldsymbol{a}(i|k\Delta t)$ to indicate a property that is predicted i-steps ahead compared to the current discrete time instant $k\Delta t$. We denote vectorial/scalar properties of SV and the inspectors with the subscripts sv and ins respectively. The Lie derivative of a general scalar function h along a vector field $\boldsymbol{f}$ is $\mathcal{L}_f h(\tilde{\boldsymbol{x}}, t) := \nabla h(\tilde{\boldsymbol{x}}, t)\boldsymbol{f}$. We define the n-dimensional identity matrix as $\boldsymbol{I}_n$ and the n-dimensional zero matrix as $\boldsymbol{O}_n$.

II. BACKGROUND

In this section, we review the fundamental nonlinear dynamics governing the relative motion of an inspector spacecraft relative to another space vehicle orbiting a planetary body. We will consider the planetary body to be the Earth without loss of generality.

A. Nonlinear Relative Dynamics

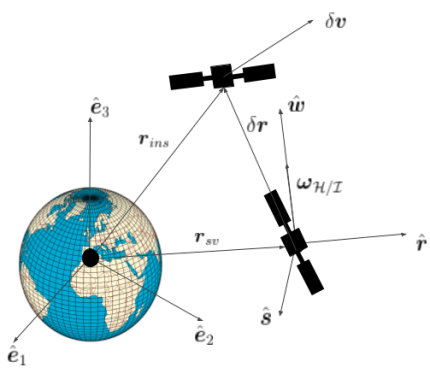

Fig. 1: Visualization of the relative state of the inspector spacecraft ($\delta\boldsymbol{r}$) with respect to the a general SV ($\boldsymbol{r}_{sv}$). The Hill's frame is defined by the base $\{\hat{\boldsymbol{r}}, \hat{\boldsymbol{s}}, \hat{\boldsymbol{w}}\}$ while the inertial frame $\mathcal{J}$ is defined by $\{\hat{\boldsymbol{e}}_1, \hat{\boldsymbol{e}}_2, \hat{\boldsymbol{e}}_3\}$.

Consider $\mathcal{J}$ to be an inertial frame fixed at the Earth's centre of mass with base $\{\hat{\boldsymbol{e}}_1,\hat{\boldsymbol{e}}_2,\hat{\boldsymbol{e}}_3\}$, such that $\hat{\boldsymbol{e}}_3$ is aligned with the rotational axis of the Earth. Furthermore, consider the Local Vertical Local Horizontal frame $\mathcal{H}$ over the SV's inertial state with base $\{\hat{\boldsymbol{r}}, \hat{\boldsymbol{s}}, \hat{\boldsymbol{w}}\}$ with $\hat{\boldsymbol{r}} = \frac{\boldsymbol{r}_{sv}}{\|\boldsymbol{r}_{sv}\|}$, $\hat{\boldsymbol{w}} = \frac{\boldsymbol{r}_{sv} \times \dot{\boldsymbol{r}}_{sv}}{\|\boldsymbol{r}_{sv} \times \dot{\boldsymbol{r}}_{sv}\|}$, $\hat{\boldsymbol{s}} = \hat{\boldsymbol{w}} \times \hat{\boldsymbol{r}}$, where $\boldsymbol{r}_{sv}$ and $\dot{\boldsymbol{r}}_{sv}$ are the inertial position and velocity of SV with respect to $\mathcal{J}$. We define the relative position and velocity of the inspector in the $\mathcal{H}$ frame as $\delta\boldsymbol{r} = \boldsymbol{r}_{ins} - \boldsymbol{r}_{sv}$ and $\delta\boldsymbol{v} = \delta\dot{\boldsymbol{r}} - \boldsymbol{\omega}_{\mathcal{H}/\mathcal{J}} \times \delta\boldsymbol{r}$ respectively, with $\boldsymbol{\omega}_{\mathcal{H}/\mathcal{J}}$ being the angular velocity of $\mathcal{H}$ w.r.t to $\mathcal{J}$ and $\delta\dot{\boldsymbol{r}}$ being the time derivative of $\delta\boldsymbol{r}$ w.r.t $\mathcal{J}$. Given the inspector relative state $\boldsymbol{x} = [\delta\boldsymbol{r}, \delta\boldsymbol{v}]^T \in \tilde{\mathbb{X}} \subset \mathbb{R}^6$ and time $t \in \mathbb{I} \subset \mathbb{R}_{\geq 0}$, the nominal dynamics of an inspector relative to the SV written in state space form is given as

$$\dot{\boldsymbol{x}} = \eta(\boldsymbol{x}, t) = \boldsymbol{f}(\boldsymbol{x}, t) + \boldsymbol{g}(\boldsymbol{x}, t)\boldsymbol{u} \tag{1}$$

with $\boldsymbol{g}(\boldsymbol{x}, t) = [\boldsymbol{O}_3, \boldsymbol{I}_3]^T$, $\boldsymbol{f}(\boldsymbol{x}, t) = [\boldsymbol{f}_r(\boldsymbol{x}, t), \boldsymbol{f}_v(\boldsymbol{x}, t)]^T$ and $\boldsymbol{u} \in \mathbb{U} \subset \mathbb{R}^3$ being the propulsive acceleration from the inspector's propulsion system. We assume $\mathbb{U}$ and $\tilde{\mathbb{X}}$ are compact sets containing the origin, and $\mathbb{I}$ is a compact time interval. The full dynamics of (1) can be found in [7, Eqs. 4.14-4.16] and it is omitted here for brevity. Nonetheless, we highlight the fact that both $\boldsymbol{f}$ and $\boldsymbol{g}$ are of class C^∞ in t and $\boldsymbol{x}$ for any compact set $\tilde{\mathbb{X}}$ such that $\delta\boldsymbol{r} \neq \boldsymbol{r}_{sv}; \forall \boldsymbol{x} \in \tilde{\mathbb{X}}$. Under the effect of orbital perturbations, the nominal dynamics in (1) becomes an input perturbed dynamics as follows

$$\dot{\tilde{\boldsymbol{x}}} = \tilde{\eta}(\tilde{\boldsymbol{x}}, t, \boldsymbol{d}) = \boldsymbol{f}(\tilde{\boldsymbol{x}}, t) + \boldsymbol{g}(\tilde{\boldsymbol{x}}, t)(\boldsymbol{u} + \boldsymbol{d}) \tag{2}$$

where $\eta(\boldsymbol{x}, t) \triangleq \tilde{\eta}(\tilde{\boldsymbol{x}}, t, \boldsymbol{0})$, $\boldsymbol{d} \in \mathbb{W} \subset \mathbb{R}^3$ is the orbital perturbation vector gathering J_2 and aerodynamic drag perturbations [13], $\tilde{\boldsymbol{x}} = [\delta\tilde{\boldsymbol{r}}, \delta\tilde{\boldsymbol{v}}] \in \tilde{\mathbb{X}} \subset \mathbb{R}^6$ is the perturbed state. The tilde notation is used to distinguish the perturbed state dynamics from the nominal dynamics. Although analytical models for $\boldsymbol{d}$ exist [13], they depend on parameters that are difficult to identify online, and therefore we consider $\boldsymbol{d}$ to be a bounded unknown disturbance that encompasses the different noise models. We further assume $\mathbb{W}$ to be a compact and convex set containing the origin such that $\mathbb{I}, \tilde{\mathbb{X}}, \mathbb{W}$ and $\mathbb{U}$ take the following form:

$$\begin{gathered} \tilde{\mathbb{X}} = \{\delta\tilde{\boldsymbol{x}} \in \mathbb{R}^6 : \|\delta\tilde{\boldsymbol{x}}\| \leq \epsilon_{\tilde{x}}\}, \mathbb{I} = [0, t_{max}], \\ \mathbb{U} = \{\boldsymbol{u} \in \mathbb{R}^3 : \|\boldsymbol{u}\| \leq \epsilon_u\}, \mathbb{W} = \{\boldsymbol{d} \in \mathbb{R}^3 : \|\boldsymbol{d}\| \leq \epsilon_d\} \end{gathered} \tag{3}$$

with $\epsilon_u, \epsilon_d, \epsilon_{\tilde{x}}, t_{max} \in \mathbb{R}_{>0}$. We will denote as $\eta_d(\boldsymbol{x}, k\Delta t)$ the discrete-time version of the nominal dynamics $\eta(\boldsymbol{x}, t)$, obtained through Runge-Kutta integration so that $\boldsymbol{x}((k+1)\Delta t) = \eta_d(\boldsymbol{x}(k\Delta t), k\Delta t)$. As we restrict our derivations to SV in nearly circular orbits, we use passive relative orbits [7, Ch. 5] as suitable reference trajectories to be tracked by each inspector. We will refer to the reference PRO trajectory state at time t as $\boldsymbol{x}_r(t) = [\delta\boldsymbol{r}_r(t), \delta\boldsymbol{v}_r(t)]$ according to the definitions in Table I.

direction	$\delta\boldsymbol{r}_r(t)$	$\delta\boldsymbol{v}_r(t)$	$\delta\boldsymbol{a}_r(t)$
$\hat{\boldsymbol{r}}$	$\rho_r \sin\omega_w^r$	$\omega_w\rho_r \cos\omega_w^r$	$-\omega_w^2\rho_r \sin\omega_w^r$
$\hat{\boldsymbol{s}}$	$\rho_s + 2\rho_r \cos\omega_w^r$	$-2\omega_w\rho_r \sin\omega_w^r$	$-2\omega_w^2\rho_r \cos\omega_w^r$
$\hat{\boldsymbol{w}}$	$\rho_w \sin\omega_w^w$	$\omega_w\rho_w \cos\omega_w^w$	$-\omega_w^2\rho_w \sin\omega_w^w$

TABLE I: Component wise definition of a PRO orbit in *amplitude-phase* parameters in the Local Vertical Local Horizontal frame $\mathcal{H}$. The shorthand notation $\omega_w^r \triangleq \omega_w t + \alpha_r$, $\omega_w^w \triangleq \omega_w t + \alpha_w$ is applied.

The parameters ρ_r,ρ_s,α_r and α_w defining a PRO are all positive scalars, and they are all design parameters. The term ω_w is the mean motion of the SV orbit and it is given as $\omega_w = \sqrt{\mu/p^3}$ [s^{-1}] with p being the semi-major axis of the SV orbit and μ being the standard gravitational parameters of the Earth. Such a reference trajectory can be proved to be an unforced solution of the Cloessy-Wiltshire (CW) model [7, Ch. 5], which is obtained by linearising

the nominal dynamics (1) around the SV inertial position and assuming the SV's orbit to be perfectly circular (zero eccentricity) and under zero orbital perturbations. Assuming an inspector is correctly deployed into a PRO, due to the effect of orbital perturbations (which are considered to be absent in the CW model) and the nonlinearity of the real dynamics, the inspector would eventually drift from its assigned PRO if not properly stabilised. We hence define the position and velocity tracking errors under nominal dynamics as $\boldsymbol{e}_{\delta\boldsymbol{r}}(t) \triangleq \delta\boldsymbol{r}(t) - \delta\boldsymbol{r}_r(t)$, $\boldsymbol{e}_{\delta\boldsymbol{v}}(t) \triangleq \delta\boldsymbol{v}(t) - \delta\boldsymbol{v}_r(t)$ and under perturbed dynamics as $\tilde{\boldsymbol{e}}_{\delta\boldsymbol{r}}(t) \triangleq \delta\tilde{\boldsymbol{r}}(t) - \delta\boldsymbol{r}_r(t)$, $\tilde{\boldsymbol{e}}_{\delta\boldsymbol{v}}(t) \triangleq \delta\tilde{\boldsymbol{v}}(t) - \delta\boldsymbol{v}_r(t)$. Additionally, $\boldsymbol{e}(t) = [\boldsymbol{e}_{\delta\boldsymbol{r}}, \boldsymbol{e}_{\delta\boldsymbol{v}}]^T$ and $\tilde{\boldsymbol{e}}(t) = [\tilde{\boldsymbol{e}}_{\delta\boldsymbol{r}}, \tilde{\boldsymbol{e}}_{\delta\boldsymbol{v}}]^T$. Without loss of generality, we will consider the reference acceleration to be tracked by each inspector as equal to zero ($\delta\boldsymbol{a}_r(t) = \boldsymbol{0}$).

B. High Order Control Barrier Functions

Control barrier functions (CBFs) [14] and their high order version (HOCBFs) [15] are a commonly applied analytical tool to define control invariant sets where the system state can evolve *safely* w.r.t a given notion of safety (*i.e.* avoid a given area where an obstacle is located or avoid a region of instability for the system). In this section we review the key application of HOCBFs in the design of safe controllers, while we recall that CBFs are only a particular case of the former. Consider a continuously differentiable scalar function $h(\tilde{\boldsymbol{x}}, t) : \mathbb{D} \times \mathbb{I} \to \mathbb{R}$, with $\mathbb{D} \subseteq \tilde{\mathbb{X}}$ being compact, and consider $\mathcal{C}_S(t)$ and $\partial\mathcal{C}_S(t)$ to be the super level set of h and its boundary, defined as $\mathcal{C}_S(t) := \{\tilde{\boldsymbol{x}} \in \mathbb{D} : h(\tilde{\boldsymbol{x}}, t) \geq 0\}$, $\partial\mathcal{C}_S(t) := \{\tilde{\boldsymbol{x}} \in \mathbb{D} : h(\tilde{\boldsymbol{x}}, t) = 0\}$. We recall the definition of relative degree as

Definition 1. *([15, Def. 5]) The relative degree of a continuously differentiable function $h : \mathbb{D} \times \mathbb{I} \to \mathbb{R}$ with respect to system equation* (2) *is the number of times it is needed to differentiate it along the dynamics of equation* (2) *until control $\boldsymbol{u}$ explicitly shows.*

In the case $h(\tilde{\boldsymbol{x}}, t)$ is of relative degree 1, then it is possible to directly define a valid input $\boldsymbol{u}$ such that $\mathcal{C}_S(t)$ is control forward invariant as described in [14]. On the other hand, if the relative degree of h is $r > 1$ for system (2), it is possible to guarantee $\mathcal{C}_S(t)$ is a control invariant set by rendering a strict subset of $\mathcal{C}_S(t)$ forward invariant as shown next. Consider the cascade of functions $H_i : \mathbb{D} \times \mathbb{I} \to \mathbb{R}$, $\forall i = 0, \dots r$ as

$$H_0(\tilde{\boldsymbol{x}}, t) = h(\tilde{\boldsymbol{x}}, t); H_i(\tilde{\boldsymbol{x}}, t) = \dot{H}_{i\text{-}1}(\tilde{\boldsymbol{x}}, t) + \alpha_{i\text{-}1}(H_{i\text{-}1}(\tilde{\boldsymbol{x}}, t)), \tag{4}$$

where $\alpha_j(t) : \mathbb{R}_{\geq 0} \to \mathbb{R}_{\geq 0}\ \forall\, j = 0, \dots r-1$ are class $\mathcal{K}$ and C^{r-j} functions. A safe set for every H_i is defined as

$$\mathcal{C}_{S_i}(t) = \{\tilde{\boldsymbol{x}} \in \mathbb{D} : H_i \geq 0\}\ \forall i = 0, \dots r. \tag{5}$$

From the definition of H_i and $\mathcal{C}_{S_i}$ the definition of HOCBF is given as follows

Definition 2. *(modified from [15, Def. 7]) Let H_i be defined as in equation* (4) *and let $\mathcal{C}_{S_i}$ be defined as in equation* (5) *$\forall i = 0, \dots r$. The function $h(\tilde{\boldsymbol{x}}, t) : \mathbb{D} \times \mathbb{I} \to \mathbb{R}$ is a High Order Barrier Function (HOCBF) for* (2) *if it is r times differentiable in $\tilde{\boldsymbol{x}}$ and t, and there exists $\alpha_j(t) : \mathbb{R}_{\geq 0} \to \mathbb{R}_{\geq 0}\ \forall\, j = 0, \dots r-1$ class $\mathcal{K}$-functions such that α_j is of class C^{r-j} and*

$$\begin{aligned} \sup_{\boldsymbol{u} \in \mathbb{U}} \Big[\frac{\partial H_{r\text{-}1}(\tilde{\boldsymbol{x}}, t)}{\partial t} + \mathcal{L}_f H_{r\text{-}1}(\tilde{\boldsymbol{x}}, t) + \mathcal{L}_g H_{r\text{-}1}(\tilde{\boldsymbol{x}}, t)\boldsymbol{u} + \\ \mathcal{L}_g H_{r\text{-}1}(\tilde{\boldsymbol{x}}, t)\boldsymbol{d} + \alpha_{r\text{-}1}(H_{r\text{-}1}(\tilde{\boldsymbol{x}}, t)) \Big] \geq 0 \end{aligned} \tag{6}$$

$\forall \tilde{\boldsymbol{x}} \in \cap_{r\text{-}1}^{0} \mathcal{C}_{S_i}(t)$.

If condition (6) is satisfied, then there exists a control input $\boldsymbol{u} \in \mathbb{U}$ that renders $\cap_{r\text{-}1}^{0} \mathcal{C}_{S_i}(t)$ forward invariant [15, Thm. 5]. For easiness of notation we define the function $\tilde{\zeta}(\tilde{\boldsymbol{x}}, \boldsymbol{u}, \delta\boldsymbol{d}, t) : \mathbb{D} \times \mathbb{U} \times \mathbb{W} \times \mathbb{I} \to \mathbb{R}$ as

$$\begin{aligned} \tilde{\zeta}(\tilde{\boldsymbol{x}}, \boldsymbol{u}, \delta\boldsymbol{d}, t) := \frac{\partial H_{r\text{-}1}(\tilde{\boldsymbol{x}}, t)}{\partial t} + \mathcal{L}_f H_{r\text{-}1}(\boldsymbol{x}, t) + \\ \mathcal{L}_g H_{r\text{-}1}(\tilde{\boldsymbol{x}}, t)(\boldsymbol{u} + \boldsymbol{d}) + \alpha_{r\text{-}1}(H_{r\text{-}1}(\tilde{\boldsymbol{x}}, t)) \end{aligned} \tag{7}$$

and forward invariance of the safe set $\cap_{r\text{-}1}^{0} \mathcal{C}_{S_i}(t)$ is enforced by ensuring that the condition $\tilde{\zeta}(\tilde{\boldsymbol{x}}, \boldsymbol{u}, \delta\boldsymbol{d}, t) \geq 0$ is met everywhere inside $\cap_{r\text{-}1}^{0} \mathcal{C}_{S_i}(t)$. Similar to the function $\eta(\boldsymbol{x}, \boldsymbol{u}, t)$, we define the function $\zeta(\boldsymbol{x}, \boldsymbol{u}, t) \triangleq \tilde{\zeta}(\tilde{\boldsymbol{x}}, \boldsymbol{u}, \boldsymbol{0}, t)$. We highlight the fact that a CBF is an HOCBF of relative degree 1, such that no distinction is further considered between these two.

Remark 1. *In this section, we assumed the class-$\mathcal{K}$ α functions appearing in the HOCBF definition to be differentiable while the original definitions assume only Lipschitz continuity. Such an assumption is necessary in the current work for defining Lemma 1 and 2 as given in the next section.*

III. PROBLEM STATEMENT

We are interested in maintaining a set of inspector spacecrafts with dynamics as in (2), ϵ-close to a set of distinct PROs (Tab. I) relative to the ISS. The PRO set is defined such that no collision between the inspectors will be possible as long as each inspector is maintained within a safe corridor from its reference. Hence, no active collision avoidance is needed. To this end, we propose a Corridor MPC (CMPC) [11] control scheme subject to two HOCBF constraints for each independent inspector: one to constraint the maximum position tracking error and the other to constraint the maximum velocity tracking error. Formally, we will be dealing with the following problem

Problem 1. *Consider a set of n inspector spacecrafts s_i $i = 1, \dots n$ in relative orbit around a SV such that each inspector moves according to the dynamics in* (2) *and SV is in a nearly circular orbit. Consider as well n distinct reference PROs $\boldsymbol{x}_{i,r}(t) = [\delta\boldsymbol{r}_{i,r}, \delta\boldsymbol{v}_{i,r}]^T$ $i = 1, \dots n$ with relative velocity $\delta\boldsymbol{v}_r(t)$ and position $\delta\boldsymbol{r}_r(t)$ according to Table I such that each inspector s_i is assigned to a specific reference PRO $\boldsymbol{x}_{r,i}$. The problem we tackle is to synthesise an appropriate ZOH feedback control input $K_i(\tilde{\boldsymbol{x}}, t) \in \mathbb{U}$ for each inspector under dynamics* (2) *such that*

$$\|\delta\tilde{\boldsymbol{r}}_i(t) - \delta\boldsymbol{r}_{i,r}(t)\| \leq \epsilon_{\delta\boldsymbol{r}}\ \forall t \in \mathbb{I}$$

with $\epsilon_{\delta \boldsymbol{r}} \in \mathbb{R}_{>0}$

IV. SAMPLED DATA HOCBF

With the end of solving Problem 1, we present our definition of sampled-data HOCBF, that will be then applied inside the control scheme presented in Section V.

It is known that for a *zero order hold* (ZOH) sampled-data system, the state measurements are only available at discrete time steps $k\Delta t$. Assuming that a control input $\boldsymbol{u}(k\Delta t) \in \mathbb{U}$ is applied to (2) such that $\tilde{\zeta}(\tilde{\boldsymbol{x}}(k\Delta t), \boldsymbol{u}(k\Delta t), \delta\boldsymbol{d}(k\Delta t), k\Delta t) \geq 0$ is satisfied, this condition is not sufficient to guarantee safety throughout the sampling interval $[k\Delta t, (k+1)\Delta t]$. In this section we present our first contribution to solve this problem by expanding the definition of sampled-data CBF developed in [11] to the HOCBF case. The following lemma is presented following tightly Lemma 1 in [11].

Lemma 1. *Consider the perturbed control affine system* (2) *where the functions* $\boldsymbol{g} : \mathbb{R}^n \times \mathbb{R}_{\geq 0} \to \mathbb{R}^{n\times m}$ *and* $\boldsymbol{f} : \mathbb{R}^n \times \mathbb{R}_{\geq 0} \to \mathbb{R}^n$ *are at least* C^{r+1} *in* $\tilde{\boldsymbol{x}}$ *and* t *on the set* $\tilde{\mathbb{X}} \times \mathbb{I}$. *Let* $\boldsymbol{d}(t) \in \mathbb{W}$ *be a bounded unknown piece-wise differentiable disturbance defined on the compact set* $\mathbb{W}$ *such that* $\|\boldsymbol{d}(t)\| \leq \epsilon_d \; \forall t \in \mathbb{I}$. *Let* $[k\Delta t, (k+1)\Delta t) \subset \mathbb{I}$ *be a sampling interval for some* $\Delta t > 0$ *such that* (2) *is subject to a constant bounded feedback control input* $\boldsymbol{u}(t) = \boldsymbol{u}(\tilde{\boldsymbol{x}}(k\Delta t), k\Delta t) \in \mathbb{U} \; \forall t \in [k\Delta t, (k+1)\Delta t]$ *shorthanded as* $\boldsymbol{u}(k\Delta t)$. *Furthermore consider a HOCBF* $h : \mathbb{D} \times \mathbb{I} \to \mathbb{R}$ *of relative degree* r *as in Def. 2 that is at least* C^{r+1} *on* $\mathbb{D} \times \mathbb{I}$ *and where* α_j *is* $C^{r-j} \; \forall j = 0, \dots r-1$. *Let* $\cap_{r\text{-}1}^{0} \mathcal{C}_{S_i}(t) \subset \mathbb{D}$ *be the associated safe sets as in* (5). *Given that at time instant* $k\Delta t$, $\tilde{\boldsymbol{x}}(k\Delta t) \in \cap_{r\text{-}1}^{0} \mathcal{C}_{S_i}(t)$, $\boldsymbol{x}(k\Delta t) \triangleq \tilde{\boldsymbol{x}}(k\Delta t)$ *and that the constant feedback control input* $\boldsymbol{u}(k\Delta t) \in \mathbb{U}$ *satisfies*

$$\zeta(\boldsymbol{x}(k\Delta t), \boldsymbol{u}(k\Delta t), k\Delta t) - L_w \Delta t - c_w \epsilon_d \geq 0 \tag{8}$$

where L_w *is defined over the set* $\mathcal{Q} \triangleq \mathbb{D} \times \mathbb{U} \times \mathbb{W} \times \mathbb{I}$

$$L_w = \max_{(\tilde{\boldsymbol{x}}, \boldsymbol{u}, \delta\boldsymbol{d}, t) \in \mathcal{Q}} |\dot{\tilde{\zeta}}(\tilde{\boldsymbol{x}}, \boldsymbol{u}, \delta\boldsymbol{d}, t)|,$$

and the constant c_w *is defined as*

$$c_w = \max_{(\tilde{\boldsymbol{x}}, t) \in \mathbb{D} \times \mathbb{I}} ||\mathcal{L}_g H_{r\text{-}1}(\tilde{\boldsymbol{x}}, t)||,$$

then it holds that

$$\tilde{\boldsymbol{x}}(k\Delta t) \in \cap_{r\text{-}1}^{0} \mathcal{C}_{S_i}(k\Delta t) \Rightarrow \tilde{\boldsymbol{x}}(t) \in \cap_{r\text{-}1}^{0} \mathcal{C}_{S_i}(t)$$

$\forall t \in [k\Delta t, (k+1)\Delta t]$.

Proof. Given a constant feedback input $\boldsymbol{u}(k\Delta t)$ on the interval $[k\Delta t, (k+1)\Delta t]$, it is assumed that both $\boldsymbol{u}(k\Delta t)$ and $\boldsymbol{d}$ are bounded on $[k\Delta t, (k+1)\Delta t)$ according to (3). Since $\boldsymbol{f}$ and $\boldsymbol{g}$ in (2) are C^{r+1} and $\boldsymbol{d}(t)$ is piece-wise differentiable, then the solution $\tilde{\boldsymbol{x}}(t)$ to (2) is uniquely defined on an interval $[k\Delta t, \tau] \subset [k\Delta t, (k+1)\Delta t)$ for some $\tau \leq (k+1)\Delta t$ [16, Thm. 54]. In addition, since $\tilde{\boldsymbol{x}}(k\Delta t) \in \cap_{r\text{-}1}^{0} \mathcal{C}_{S_i}(t) \subset \mathbb{D}$ and by continuity of $\tilde{\boldsymbol{x}}(t)$ there exist $\tau_0 \in [k\Delta t, \tau]$ such that $\tilde{\boldsymbol{x}}(t) \in \mathbb{D} \; \forall t \in [k\Delta t, \tau_0]$. The function $\tilde{\zeta}$ on $[k\Delta t, \tau_0]$ is then written as functions of time as

$$\begin{aligned} \tilde{\zeta}(\tilde{\boldsymbol{x}}(t), \boldsymbol{u}(k\Delta t), \boldsymbol{d}(t), t) = \frac{\partial H_{r\text{-}1}(\tilde{\boldsymbol{x}}(t), t)}{\partial t} + \\ \mathcal{L}_f H_{r\text{-}1}(\tilde{\boldsymbol{x}}(t), t) + \mathcal{L}_g H_{r\text{-}1}(\tilde{\boldsymbol{x}}(t), t)(\boldsymbol{u}(k\Delta t) + \boldsymbol{d}(t)) + \\ \alpha_{r\text{-}1}(H_{r\text{-}1}(\tilde{\boldsymbol{x}}(t), t)) \quad \forall t \in [k\Delta t, \tau_0]. \end{aligned}$$

It is noted that $\tilde{\zeta}(\tilde{\boldsymbol{x}}(t), \boldsymbol{u}(k\Delta t), \boldsymbol{d}(t), t)$ is a piece-wise differentiable function of time as all the functions appearing in its definition are continuously differentiable apart from $\boldsymbol{d}(t)$, which is only piece-wise differentiable. Since this function respects the conditions in [17, Prop 4.1.2], the Lipschitz constant for $\tilde{\zeta}(\tilde{\boldsymbol{x}}(t), \boldsymbol{u}(k\Delta t), \boldsymbol{d}(t), t)$ exists and it is given by

$$L_w = \max_{(\tilde{\boldsymbol{x}}, \boldsymbol{u}, \delta\boldsymbol{d}, t) \in \mathcal{Q}} |\dot{\tilde{\zeta}}(\tilde{\boldsymbol{x}}, \boldsymbol{u}(k\Delta t), \delta\boldsymbol{d}, t)|.$$

After introducing the shorthand notation $\tilde{\zeta}_k(\tilde{x}(t)) \triangleq \tilde{\zeta}(\tilde{\boldsymbol{x}}(t), \boldsymbol{u}(k\Delta t), \boldsymbol{d}(t), t)$ and $\zeta_k(\boldsymbol{x}(t)) \triangleq \zeta(\boldsymbol{x}(t), \boldsymbol{u}(k\Delta t), t)$ we can apply the Lipschitz continuity property on $\tilde{\zeta}_k(\tilde{\boldsymbol{x}}(t))$ such that $|\tilde{\zeta}_k(\tilde{\boldsymbol{x}}(t_2)) - \tilde{\zeta}_k(\tilde{\boldsymbol{x}}(t_1))| \leq L_w |t_2 - t_1| \; \forall t_1, t_2 \in [k\Delta t, \tau_0]$. The maximum negative variation of $\tilde{\zeta}_k(\tilde{\boldsymbol{x}}(t))$ over the interval $[k\Delta t, \tau_0]$ then satisfies

$$\tilde{\zeta}_k(\tilde{\boldsymbol{x}}(t_2)) - \tilde{\zeta}_k(\tilde{\boldsymbol{x}}(t_1)) \geq -L_w |t_2 - t_1| \; \forall t_1, t_2 \in [k\Delta t, \tau_0]. \tag{9}$$

Recalling that at time $k\Delta t$, $\tilde{\boldsymbol{x}}(k\Delta t) = \boldsymbol{x}(k\Delta t)$ the following relation between $\tilde{\zeta}_k(\tilde{\boldsymbol{x}}(k\Delta t))$ and $\zeta_k(\boldsymbol{x}(k\Delta t))$ holds

$$\begin{aligned} &\tilde{\zeta}_k(\tilde{\boldsymbol{x}}(k\Delta t)) = \\ &\quad \zeta_k(\boldsymbol{x}(k\Delta t)) + \mathcal{L}_g H_{r\text{-}1}(\tilde{\boldsymbol{x}}(k\Delta t), k\Delta t)\boldsymbol{d}(k\Delta t) \geq \\ &\quad \zeta_k(\boldsymbol{x}(k\Delta t)) - \max_{(\tilde{\boldsymbol{x}}, t) \in \mathbb{D} \times \mathbb{I}} ||\mathcal{L}_g H_{r\text{-}1}(\tilde{\boldsymbol{x}}, t)|| \, ||\boldsymbol{d}|| = \\ &\quad \zeta_k(\boldsymbol{x}(k\Delta t)) - c_w \epsilon_d. \end{aligned} \tag{10}$$

By replacing t_1 with $k\Delta t$, t_2 with t in (9) and by adding and subtracting $\tilde{\zeta}_k(\tilde{\boldsymbol{x}}(k\Delta t))$ on the LHS and RHS of (9) we obtain

$$\begin{aligned} &\tilde{\zeta}_k(\tilde{\boldsymbol{x}}(t)) \geq -L_w \Delta t + \tilde{\zeta}_k(\tilde{\boldsymbol{x}}(k\Delta t)) \\ &\quad \geq -L_w \Delta t + \zeta_k(\boldsymbol{x}(k\Delta t)) - c_w \epsilon_d \; \forall t \in [k\Delta t, \tau_0]. \end{aligned} \tag{11}$$

Replacing (8) from the lemma statement in (11) it is now evident that $\tilde{\zeta}_k(\tilde{\boldsymbol{x}}(t)) \geq 0 \; \forall t \in [k\Delta t, \tau_0]$ which by [15, Thm. 5] ensures that $\tilde{\boldsymbol{x}}(t) \in \cap_{r\text{-}1}^{0} \mathcal{C}_{S_i}(t) \; \forall t \in [k\Delta t, \tau_0]$. We will now prove that $\tilde{\boldsymbol{x}}(t) \in \cap_{r\text{-}1}^{0} \mathcal{C}_{S_i}(t) \; \forall t \in [k\Delta t, \tau]$ by contradiction. Suppose instead that for some $\tau_a \in (\tau_0, \tau]$, $\tilde{\boldsymbol{x}}(\tau_a) \in \mathbb{D} \setminus \cap_{r\text{-}1}^{0} \mathcal{C}_{S_i}(t)$ and $\tilde{\boldsymbol{x}}(t) \in \mathbb{D}$ for all $t \in [k\Delta t, \tau_a]$ (i.e., the solution has left $\cap_{r\text{-}1}^{0} \mathcal{C}_{S_i}(t)$, but not $\mathbb{D}$). Then $\tilde{\boldsymbol{x}}(t)$ must leave $\cap_{r\text{-}1}^{0} \mathcal{C}_{S_i}(t)$ at some $t < \tau_a$. Furthermore, since the closed-loop dynamics are differentiable on $\mathbb{D}$, $\tilde{\boldsymbol{x}}(t)$ is uniquely defined on $[k\Delta t, \tau_a]$ (this is shown by repeatedly applying [16, Thm. 54], since $\tilde{\boldsymbol{x}}(t)$ remains in $\mathbb{D}$ over which local differentiability of the closed-loop dynamics holds). To leave $\cap_{r\text{-}1}^{0} \mathcal{C}_{S_i}(t)$, $\dot{H}_{r\text{-}1}(\tilde{\boldsymbol{x}}, t) < 0$ must hold on $\partial(\cap_{r\text{-}1}^{0} \mathcal{C}_{S_i}(t))$. The maximum negative variation of $\tilde{\zeta}_k(\tilde{\boldsymbol{x}}(t))$ is then recomputed over the interval $[k\Delta t, \tau_a]$ and it is obtained again that $\tilde{\zeta}_k(\tilde{\boldsymbol{x}}(t)) \geq 0 \; \forall t \in [k\Delta t, \tau_a]$ as L_w and c_w are independent from τ_a as well as τ_0. Therefore we see that $\dot{H}_{r\text{-}1}(\tilde{\boldsymbol{x}}, t) \geq 0$ holds for any $\tilde{\boldsymbol{x}}(t) \in \cap_{r\text{-}1}^{0} \mathcal{C}_{S_i}(t), t \in$

$[k\Delta t, \tau_a]$ such that $\tau_a \leq (k+1)\Delta t$. Hence, the contradiction is reached, and so $\tilde{\boldsymbol{x}}(t)$ can never leave $\cap_{r\text{-}1}^{0}\mathcal{C}_{S_i}(t)$ (and thus $\mathbb{D}$) on $t \in [k\Delta t, \tau]$ with $\tau \leq (k+1)\Delta t$. Since it was showed that $\tilde{\boldsymbol{x}}(t)$ remains in a compact subset on the interval $[k\Delta t, \tau]$ (namely $\mathbb{D}$), then $\tilde{\boldsymbol{x}}(t)$ exist and is unique over the whole interval $[k\Delta t, (k+1)\Delta t]$ [16, Prop. C.3.6]. By the same arguments applied for the previous sub-intervals, it is proved again that $\tilde{\zeta}_k(\tilde{\boldsymbol{x}}(t)) \geq 0 \; \forall t \in [k\Delta t, (k+1)\Delta t]$ which ensures that $\tilde{\boldsymbol{x}}(t) \in \cap_{r\text{-}1}^{0}\mathcal{C}_{S_i}(t) \; \forall t \in [k\Delta t, (k+1)\Delta t]$ by [15, Thm.5]. □

As already noticed in [10], satisfying (8) might require excessive control authority given c_w,L_w are computed on $\mathcal{Q}$. For this reason, we introduce the definition of reachable set, where locally valid parameters L_w^l and c_w^l can be defined, as follows:

Definition 3. *Given a set $\mathcal{N} \subseteq \tilde{\mathbb{X}}$ we define ${}^{\Delta t}\mathcal{R}(\mathcal{N})$ as the set of all states $\tilde{\boldsymbol{x}} \in \tilde{\mathbb{X}}$ that can be reached from $\tilde{\boldsymbol{x}} \in \mathcal{N}$ in a time interval Δt with state dynamics as in* (2) *and under available control input $\boldsymbol{u} \in \mathbb{U}$ according to* (3)*.*

The following Lemma is proposed as a less conservative modification of Lemma 1.

Lemma 2. *Consider the perturbed control affine system* (2) *where the functions $\boldsymbol{g} : \mathbb{R}^n \times \mathbb{R}_{\geq 0} \to \mathbb{R}^{n\times m}$ and $\boldsymbol{f} : \mathbb{R}^n \times \mathbb{R}_{\geq 0} \to \mathbb{R}^n$ are at least C^{r+1} in $\tilde{\boldsymbol{x}}$ and t on the set $\tilde{\mathbb{X}} \times \mathbb{I}$. Let $\boldsymbol{d}(t) \in \mathbb{W}$ be a bounded unknown piecewise differentiable disturbance defined on the compact set $\mathbb{W}$ such that $\|\boldsymbol{d}(t)\| \leq \epsilon_d \; \forall t \in \mathbb{I}$. Let $[k\Delta t, (k+1)\Delta t] \subset \mathbb{I}$ be a sampling interval for some $\Delta t > 0$ such that* (2) *is subject to a constant bounded feedback control input $\boldsymbol{u}(t) = \boldsymbol{u}(\tilde{\boldsymbol{x}}(k\Delta t), k\Delta t) \in \mathbb{U} \; \forall t \in [k\Delta t, (k+1)\Delta t]$ shorthanded as $\boldsymbol{u}(k\Delta t)$. Furthermore consider a HOCBF $h : \mathbb{D} \times \mathbb{I} \to \mathbb{R}$ of relative degree r as in Def. 2 that is at least C^{r+1} on $\mathbb{D} \times \mathbb{I}$ and where α_j is $C^{r-j} \; \forall j = 0, \dots r-1$. Let $\cap_{r\text{-}1}^{0}\mathcal{C}_{S_i}(t) \subset \mathbb{D}$ be the associated safe set as in* (5)*. Given that at time instant $k\Delta t$, $\tilde{\boldsymbol{x}}(k\Delta t) \in \cap_{r\text{-}1}^{0}\mathcal{C}_{S_i}(t)$, $\boldsymbol{x}(k\Delta t) \triangleq \tilde{\boldsymbol{x}}(k\Delta t)$ and that the constant feedback control input $\boldsymbol{u}(k\Delta t) \in \mathbb{U}$ respects the condition*

$$\begin{aligned} &\zeta(\boldsymbol{x}(k\Delta t), \boldsymbol{u}(k\Delta t), k\Delta t) - \\ &\quad L_w^l(\tilde{\boldsymbol{x}}(k\Delta t), \Delta t)\Delta t - c_w^l(\tilde{\boldsymbol{x}}(k\Delta t), \Delta t)\epsilon_d \geq 0 \end{aligned} \tag{12}$$

where $L_w^l(\tilde{\boldsymbol{x}}(k\Delta t), \Delta t)$ is defined over the set ${}^{\Delta t}\mathcal{Q}(\cap_{r\text{-}1}^{0}\mathcal{C}_{S_i}(t)) \triangleq {}^{\Delta t}\mathcal{R}(\cap_{r\text{-}1}^{0}\mathcal{C}_{S_i}(t)) \times \mathbb{U} \times \mathbb{W} \times [k\Delta t, (k+1)\Delta t]$ as

$$L_w^l(\tilde{\boldsymbol{x}}(k\Delta t), \Delta t) = \max_{\substack{(\tilde{\boldsymbol{x}}, \boldsymbol{u}, \boldsymbol{d}, t) \in \\ {}^{\Delta t}\mathcal{Q}(\cap_{r\text{-}1}^{0}\mathcal{C}_{S_i}(k\Delta t))}} |\dot{\tilde{\zeta}}(\tilde{\boldsymbol{x}}, \boldsymbol{u}(k\Delta t), \delta\boldsymbol{d}, t)|,$$

and the constant $c_w^l(\tilde{\boldsymbol{x}}(k\Delta t), \Delta t)$ is defined as

$$c_w^l(\tilde{\boldsymbol{x}}(k\Delta t), \Delta t) = \max_{{}^{\Delta t}\mathcal{R}(\cap_{r\text{-}1}^{0}\mathcal{C}_{S_i}(k\Delta t)) \times \mathbb{I}} ||\mathcal{L}_g \mathcal{L}_f^{r-1} h(\tilde{\boldsymbol{x}}, t)||.$$

Then it holds that

$$\tilde{\boldsymbol{x}}(k\Delta t) \in \cap_{r\text{-}1}^{0}\mathcal{C}_{S_i}(k\Delta t) \Rightarrow \tilde{\boldsymbol{x}}(t) \in \cap_{r\text{-}1}^{0}\mathcal{C}_{S_i}(t) \tag{13}$$

$\forall t \in [k\Delta t, (k+1)\Delta t]$.

Proof. The proof of the lemma follows directly from the proof of Lemma 1 by simply noting that inside the interval $[k\Delta t, (k+1)\Delta t]$ all the solutions $\tilde{\boldsymbol{x}}(t)$ to (2) evolve inside ${}^{\Delta t}\mathcal{R}(\cap_{r\text{-}1}^{0}\mathcal{C}_{S_i}(k\Delta t))$. □

Based on Lemma 2, we are now ready to give our definition of sampled-data HOCBF (SD-HOCBF).

Definition 4 (SD-HOCBF)**.** *Consider a HOCBF $h : \mathbb{D} \times \mathbb{I} \to \mathbb{R}$ as in Definition 2 with relative degree r with respect to* (2) *and corresponding safe set $\cap_{r\text{-}1}^{0}\mathcal{C}_{S_i}(t)$ according to* (5)*. The function h is a sampled-data HOCBF (SD-HOCBF) for a given $\Delta t > 0$ if for any point $\tilde{\boldsymbol{x}} \in \cap_{r\text{-}1}^{0}\mathcal{C}_{S_i}(k\Delta t)$ and $k \in \mathbb{N}_0$, there is a constant feedback input $\boldsymbol{u}(\boldsymbol{x}(k\Delta t), k\Delta t) \in \mathbb{U}$, shortened to $\boldsymbol{u}(k\Delta t)$, where $\boldsymbol{x}(k\Delta t) \triangleq \tilde{\boldsymbol{x}}(k\Delta t)$, such that:*

$$\begin{aligned} &\sup_{\boldsymbol{u}\in\mathbb{U}}[\zeta(\boldsymbol{x}(k\Delta t), \boldsymbol{u}(\Delta t), k\Delta t) - \\ &\quad L_w^l(\tilde{\boldsymbol{x}}(k\Delta t), \Delta t)\Delta t - c_w^l(\tilde{\boldsymbol{x}}(k\Delta t), \Delta t)\epsilon_d] \geq 0 \end{aligned} \tag{14}$$

where $L_w^l(\tilde{\boldsymbol{x}}(k\Delta t), \Delta t)$ and $c_w^l(\tilde{\boldsymbol{x}}(k\Delta t), \Delta t)$ are defined according to Lemma.2

V. CONTROL STRATEGY

In this section we present the CMPC scheme applied to solve Problem 1 leveraging the newly introduced definition of SD-HOCBF. We first introduce the two SD-HOCBF that will be included in the CMPC scheme :

$$h_{\delta\boldsymbol{v}}(\tilde{\boldsymbol{x}}, t) = \epsilon_{\delta\boldsymbol{v}}^2 - \|\delta\tilde{\boldsymbol{v}} - \delta\boldsymbol{v}_r(t)\|^2 = \epsilon_{\delta\boldsymbol{v}}^2 - \|\tilde{\boldsymbol{e}}_{\delta\boldsymbol{v}}(t)\|^2 \tag{15a}$$

$$h_{\delta\boldsymbol{r}}(\tilde{\boldsymbol{x}}, t) = \epsilon_{\delta\boldsymbol{r}}^2 - \|\delta\tilde{\boldsymbol{r}} - \delta\boldsymbol{r}_r(t)\|^2 = \epsilon_{\delta\boldsymbol{r}}^2 - \|\tilde{\boldsymbol{e}}_{\delta\boldsymbol{r}}(t)\|^2, \tag{15b}$$

where $h_{\delta\boldsymbol{v}}(\tilde{\boldsymbol{x}}, t)$ and $h_{\delta\boldsymbol{r}}(\tilde{\boldsymbol{x}}, t)$ are applied to bound the maximum velocity error norm and position error norm respectively. While $h_{\delta\boldsymbol{v}}(\tilde{\boldsymbol{x}}, t)$ is directly applied to tackle Problem 1, the barrier $h_{\delta\boldsymbol{r}}(\tilde{\boldsymbol{x}}, t)$ is applied to provide recursive feasibility guarantees on the satisfaction of the former as it will be developed in this section.

It can be shown that $h_{\delta\boldsymbol{r}}(\tilde{\boldsymbol{x}}, t)$ is a relative degree two HOCBF for (2) while $h_{\delta\boldsymbol{v}}(\tilde{\boldsymbol{x}}, t)$ is of relative degree one. Given the class-$\mathcal{K}$ functions $\alpha_{\delta\boldsymbol{r}_0}(x) = p_{\delta\boldsymbol{r}_0}x$, $\alpha_{\delta\boldsymbol{r}_1}(x) = p_{\delta\boldsymbol{r}_1}x$ for $h_{\delta\boldsymbol{r}}$ and $\alpha_{\delta\boldsymbol{v}_0}(x) = p_{\delta\boldsymbol{v}_0}x$ for $h_{\delta\boldsymbol{v}}$, with $p_{\delta\boldsymbol{r}_0}, p_{\delta\boldsymbol{r}_1}, p_{\delta\boldsymbol{v}_0} \in \mathbb{R}_{>0}$, we can define the functions $\tilde{\zeta}_{\delta\boldsymbol{r}}$ and $\tilde{\zeta}_{\delta\boldsymbol{v}}$ as

$$\tilde{\zeta}_{\delta\boldsymbol{v}} = -2\tilde{\boldsymbol{e}}_{\delta\boldsymbol{v}}^T(\boldsymbol{f}_v + \boldsymbol{u} + \boldsymbol{d}) + p_{\delta\boldsymbol{v}_0}(\epsilon_{\delta\boldsymbol{v}}^2 - \|\tilde{\boldsymbol{e}}_{\delta\boldsymbol{v}}\|^2) \tag{16a}$$

$$\begin{aligned} &\tilde{\zeta}_{\delta\boldsymbol{r}} = -2\|\tilde{\boldsymbol{e}}_{\delta\boldsymbol{v}}\|^2 - 2\tilde{\boldsymbol{e}}_{\delta\boldsymbol{r}}^T(\boldsymbol{f}_v + \boldsymbol{d} + \boldsymbol{u}) - \\ &2(p_{\delta\boldsymbol{r}_0} + p_{\delta\boldsymbol{r}_1})(\tilde{\boldsymbol{e}}_{\delta\boldsymbol{r}}^T\tilde{\boldsymbol{e}}_{\delta\boldsymbol{v}}) + p_{\delta\boldsymbol{r}_0}p_{\delta\boldsymbol{r}_1}(\epsilon_{\delta\boldsymbol{r}}^2 - \|\tilde{\boldsymbol{e}}_{\delta\boldsymbol{r}}\|^2) \end{aligned} \tag{16b}$$

and the safe sets definitions are then given by $\mathcal{C}_{S,\delta\boldsymbol{r}}(t) = \{\tilde{\boldsymbol{x}} \in \tilde{\mathbb{X}} : H_{\delta\boldsymbol{r}_0}(\tilde{\boldsymbol{x}}, t) \geq 0 \wedge H_{\delta\boldsymbol{r}_1}(\tilde{\boldsymbol{x}}, t) \geq 0\}, \mathcal{C}_{S,\delta\boldsymbol{v}}(t) = \{\tilde{\boldsymbol{x}} \in \tilde{\mathbb{X}} : H_{\delta\boldsymbol{v}_0}(\tilde{\boldsymbol{x}}, t) \geq 0\}, \mathcal{C}_{S,x}(t) = \mathcal{C}_{S,\delta\boldsymbol{v}}(t) \cup \mathcal{C}_{S,\delta\boldsymbol{r}}(t)\}$; with $H_{\delta\boldsymbol{r}_1}$,$H_{\delta\boldsymbol{r}_0}$,$H_{\delta\boldsymbol{v}_0}$ defined according to (4). It is evident from (16b) that the velocity error norm $\|\tilde{\boldsymbol{e}}_{\delta\boldsymbol{v}}\|$ can possibly grow unbounded inside the safe set if not bounded by the $h_{\delta\boldsymbol{v}}(\tilde{\boldsymbol{x}}, t)$. As $\tilde{\zeta}_{\delta\boldsymbol{r}}$ (and $\zeta_{\delta\boldsymbol{r}}$) depends on the negative norm square of the velocity tracking error, it is crucial to ensure that such term remains bounded to analyze the satisfaction of (14). In the remainder, we will refer to $L_w^l(\tilde{\boldsymbol{x}}, t)$ and

$c_w^l(\tilde{\boldsymbol{x}},t)$ from Lemma 2 as computed for $h_{\delta\boldsymbol{r}}$ and $h_{\delta\boldsymbol{v}}$ with the symbols $L^l_{\delta\boldsymbol{r}}$,$c^l_{\delta\boldsymbol{r}}$ and $L^l_{\delta\boldsymbol{v}}$,$c^l_{\delta\boldsymbol{v}}$ respectively.

Computing $L^l_w(\tilde{\boldsymbol{x}},\Delta t)$ and $c^l_w(\tilde{\boldsymbol{x}},\Delta t)$ requires the online computation of ${}^{\Delta t}\mathcal{R}(\mathcal{C}_{S,x}(t))$. While an analytical definition is impractical, we next propose to find a set ${}^{\Delta t}\bar{\mathcal{R}}(\mathcal{C}_{S,x}(t)) \supset {}^{\Delta t}\mathcal{R}(\mathcal{C}_{S,x}(t))$ such that suitable constants $\bar{L}^l_w(\Delta t) \geq L^l_w(\tilde{\boldsymbol{x}},\Delta t)$, $\bar{c}^l_w(\Delta t) \geq c^l_w(\tilde{\boldsymbol{x}},\Delta t)$ can be computed only based on the current sampling time Δt. This result is achieved in two steps: i) we first derive a constant $\bar{a}$ that upper bounds the maximum total acceleration that (2) is subject to; and ii) based on $\bar{a}$ we compute the maximum position and velocity tracking errors that can be reached by (2) in a sampling interval $[k\Delta t,(k+1)\Delta t]$. Namely, we compute the maximum acceleration that (2) is subject as

$$\max_{(\boldsymbol{x},t)\in\tilde{\mathbb{X}}\times\mathbb{I}} \|\delta\dot{\tilde{\boldsymbol{v}}}\| = \|\boldsymbol{f}_v + \boldsymbol{u} + \boldsymbol{d}\| \leq \epsilon_f + \epsilon_u + \epsilon_d = \bar{a} \quad (17)$$

with $\epsilon_f = \max_{(\tilde{\boldsymbol{x}},t)\in\tilde{\mathbb{X}}\times\mathbb{I}} \|\boldsymbol{f}_v(\tilde{\boldsymbol{x}},t)\|$. Note that in (17) the function $\boldsymbol{g}$ was omitted as it is the identity matrix. Next, the maximum velocity and position tracking errors are computed assuming that (2) is forced under constant acceleration $\bar{a}$ during the sampling interval $[k\Delta t,(k+1)\Delta t]$. Under these assumptions, we compute the maximum velocity and position tracking errors in first-order approximation for the time interval $[k\Delta t,(k+1)\Delta t]$ as $\|\tilde{\boldsymbol{e}}_{\delta\boldsymbol{v}}(t) - \tilde{\boldsymbol{e}}_{\delta\boldsymbol{v}}(k\Delta t)\| \leq \|\delta\tilde{\boldsymbol{v}}(t) - \delta\tilde{\boldsymbol{v}}(k\Delta t)\| + \|\delta\boldsymbol{v}_r(t) - \delta\boldsymbol{v}_r(k\Delta t)\| \leq \bar{a}\Delta t + \bar{a}_r\Delta t \;\; \forall t \in [k\Delta t,(k+1)\Delta t]$ and $\|\tilde{\boldsymbol{e}}_{\delta\boldsymbol{r}}(t) - \tilde{\boldsymbol{e}}_{\delta\boldsymbol{r}}(k\Delta t)\| \leq \|\delta\tilde{\boldsymbol{r}}(t) - \delta\tilde{\boldsymbol{r}}(k\Delta t)\| + \|\delta\boldsymbol{r}_r(t) - \delta\boldsymbol{r}_r(k\Delta t)\| \leq \bar{a}\frac{\Delta t^2}{2} + \epsilon_{\delta\boldsymbol{v}}\Delta t + \bar{a}_r\frac{\Delta t^2}{2} + \bar{v}_r\Delta t \;\; \forall t \in [k\Delta t,(k+1)\Delta t]$; where $\bar{v}_r \triangleq \max_{t\in\mathbb{I}}\|\delta\boldsymbol{v}_r(t)\|$ and $\bar{a}_r \triangleq \max_{t\in\mathbb{I}}\|\delta\boldsymbol{a}_r(t)\|$ (Tab. I). Note that we applied the formulas for linear displacement under constant acceleration to compute the maximum velocity and position error bounds. Moreover, the constants $\bar{a}$ and $\bar{v}$ exist as a PRO is a periodic trajectory. For completeness, we define $\bar{r}_r \triangleq \max_{t\in\mathbb{I}}\|\delta\boldsymbol{r}_r(t)\|$. If we define two new functions $\gamma_{\delta\boldsymbol{v}} = \bar{\epsilon}^2_{\delta\boldsymbol{v}} - \|\tilde{\boldsymbol{e}}_{\delta\boldsymbol{v}}\|^2$ and $\gamma_{\delta\boldsymbol{r}} = \bar{\epsilon}^2_{\delta\boldsymbol{r}} - \|\tilde{\boldsymbol{e}}_{\delta\boldsymbol{r}}\|^2$ with $\bar{\epsilon}_{\delta\boldsymbol{v}}(\Delta t) \triangleq \epsilon_{\delta\boldsymbol{v}} + \bar{a}\Delta t + \bar{a}_r\Delta t$ and $\bar{\epsilon}_{\delta\boldsymbol{r}}(\Delta t) \triangleq \epsilon_{\delta\boldsymbol{r}} + \bar{a}\frac{\Delta t^2}{2} + \epsilon_{\delta\boldsymbol{v}}\Delta t + \bar{a}_r\frac{\Delta t^2}{2} + \bar{v}_r\Delta t$; we can then define ${}^{\Delta t}\bar{\mathcal{R}}(\mathcal{C}_{S,x}(t))$ as ${}^{\Delta t}\bar{\mathcal{R}}(\mathcal{C}_{S,x}(t)) = \{\tilde{\boldsymbol{x}} \in \tilde{\mathbb{X}} : \gamma_{\delta\boldsymbol{v}}(\tilde{\boldsymbol{x}},t) \geq 0 \wedge \gamma_{\delta\boldsymbol{r}}(\tilde{\boldsymbol{x}},t) \geq 0\}$. We next show how we compute valid constants $\bar{L}^l_w(\Delta t) \geq L^l_w(\tilde{\boldsymbol{x}},\Delta t)$ and $\bar{c}^l_w(\Delta t) \geq c^l_w(\tilde{\boldsymbol{x}},\Delta t)$ over ${}^{\Delta t}\bar{\mathcal{R}}(\mathcal{C}_{S,x}(t))$ such that the conditions for Lemma 2 are still satisfied by $\bar{L}_w$ and $\bar{c}_w$. The time derivative of $\tilde{\zeta}_{\delta\boldsymbol{r}}$ and $\tilde{\zeta}_{\delta\boldsymbol{v}}$ is derived as $\dot{\tilde{\zeta}}_{\delta\boldsymbol{v}} = -2\tilde{\boldsymbol{e}}^T_{\delta\boldsymbol{v}}\left(\dot{\boldsymbol{f}}_v + \dot{\boldsymbol{d}}\right) - 2\|\boldsymbol{f}_v + \boldsymbol{u} + \boldsymbol{d}\|^2 - 2p_{\delta\boldsymbol{v}_0}\tilde{\boldsymbol{e}}^T_v(\boldsymbol{f}_v + \boldsymbol{u} + \boldsymbol{d})$ and $\dot{\tilde{\zeta}}_{\delta\boldsymbol{r}} = -6\tilde{\boldsymbol{e}}^T_{\delta\boldsymbol{v}}(\boldsymbol{f}_v + \boldsymbol{u} + \boldsymbol{d}) - 2\tilde{\boldsymbol{e}}^T_{\delta\boldsymbol{r}}(\dot{\boldsymbol{f}}_v + \dot{\boldsymbol{d}}) - 2(p_{\delta\boldsymbol{r}_0} + p_{\delta\boldsymbol{r}_1})(\tilde{\boldsymbol{e}}^T_{\delta\boldsymbol{v}}\tilde{\boldsymbol{e}}_{\delta\boldsymbol{v}} + \tilde{\boldsymbol{e}}^T_{\delta\boldsymbol{r}}(\boldsymbol{f}_v + \boldsymbol{u} + \boldsymbol{d})) - p_{\delta\boldsymbol{r}_0}p_{\delta\boldsymbol{r}_1}(2\tilde{\boldsymbol{e}}^T_{\delta\boldsymbol{v}}\tilde{\boldsymbol{e}}_{\delta\boldsymbol{r}})$. Noting that $\mathcal{L}_gH_{\delta\boldsymbol{v}_0}(\tilde{\boldsymbol{x}},t) = -2\tilde{\boldsymbol{e}}_{\delta\boldsymbol{v}}$ and $\mathcal{L}_gH_{\delta\boldsymbol{r}_1}(\tilde{\boldsymbol{x}},t) = -2\tilde{\boldsymbol{e}}_{\delta\boldsymbol{r}}$; we can obtain $\bar{L}^l_{\delta\boldsymbol{r}}(\Delta t)$,$\bar{L}^l_{\delta\boldsymbol{v}}(\Delta t)$,$\bar{c}^l_{\delta\boldsymbol{r}}(\Delta t)$,$\bar{c}^l_{\delta\boldsymbol{v}}(\Delta t)$ by applying the inequality $\boldsymbol{a}^T\boldsymbol{b} \leq \|\boldsymbol{a}\|\|\boldsymbol{b}\|$ in $\dot{\tilde{\zeta}}_{\delta\boldsymbol{r}}$ and $\dot{\tilde{\zeta}}_{\delta\boldsymbol{v}}$, which yields

$$\bar{L}^l_{\delta v}(\Delta t) = 2\epsilon_{\delta\boldsymbol{v}}\beta + 2\bar{a}^2 + 2p_{\delta\boldsymbol{v}_0}\bar{\epsilon}_{\delta\boldsymbol{v}}\bar{a} \geq L^l_{\delta v}(\tilde{\boldsymbol{x}},\Delta t) \quad (18a)$$

$$\bar{c}^l_{\delta v}(\Delta t) = 2\bar{\epsilon}_{\delta\boldsymbol{v}} \geq c^l_{\delta v}(\tilde{\boldsymbol{x}},\Delta t) \quad (18b)$$

$$\bar{L}^l_{\delta\boldsymbol{r}}(\Delta t) = 6\bar{a} + 2(p_{\delta\boldsymbol{r}_0} + p_{\delta\boldsymbol{r}_1})(\bar{\epsilon}^2_{\delta\boldsymbol{v}} + \bar{\epsilon}_{\delta\boldsymbol{r}}\bar{a}) + 2\epsilon_{\delta\boldsymbol{r}}\beta + 2p_{\delta\boldsymbol{r}_0}p_{\delta\boldsymbol{r}_1}(\bar{\epsilon}_{\delta\boldsymbol{r}}\bar{\epsilon}_{\delta\boldsymbol{v}}) \geq L^l_{\delta\boldsymbol{r}}(\tilde{\boldsymbol{x}},\Delta t) \quad (18c)$$

$$\bar{c}^l_{\delta\boldsymbol{v}}(\Delta t) = 2\bar{\epsilon}_{\delta\boldsymbol{r}} \geq c^l_{\delta\boldsymbol{v}}(\tilde{\boldsymbol{x}},\Delta t), \forall\tilde{\boldsymbol{x}} \in \mathcal{C}_{S,x}(t) \quad (18d)$$

where $\beta = \max_{{}^{\Delta t}\mathcal{Q}(\cap^0_{r=1}\mathcal{C}_{S_i}(t))} \|\dot{\boldsymbol{f}}_v\| + \max_{\tilde{\mathbb{X}}\times\mathbb{I}}\|\dot{\boldsymbol{d}}\|$.

A. Corridor MPC

Now that $\bar{L}^l_{\delta\boldsymbol{r}}(\Delta t)$,$\bar{L}^l_{\delta\boldsymbol{v}}(\Delta t)$,$\bar{c}^l_{\delta\boldsymbol{r}}(\Delta t)$,$\bar{c}^l_{\delta\boldsymbol{v}}(\Delta t)$ are defined, the CMPC scheme applied to control each inspector along its trajectory is presented [11, Eqs. 17-18].

$$J^*_N(\tilde{\boldsymbol{e}}_x(k\Delta t)) = \min_{\mathbf{u}^*_k} J_N(\boldsymbol{e}_x,\boldsymbol{u}) \quad (19a)$$

$$\boldsymbol{x}((m+1)|k\Delta t) = \eta_d(\boldsymbol{x}(m|k\Delta t),\boldsymbol{u}(m|k\Delta t),k\Delta t) \quad (19b)$$

$$\zeta_{\delta\boldsymbol{v}}(\boldsymbol{x}(0|k\Delta t),\boldsymbol{u}(0|k\Delta t),0) - \bar{L}^l_{\delta\boldsymbol{v}}\Delta t - \bar{c}^l_{\delta\boldsymbol{v}}\epsilon_d \geq 0 \quad (19c)$$

$$\zeta_{\delta\boldsymbol{r}}(\boldsymbol{x}(0|k\Delta t),\boldsymbol{u}(0|k\Delta t),0) - \bar{L}^l_{\delta\boldsymbol{r}}\Delta t - \bar{c}^l_{\delta\boldsymbol{r}}\epsilon_d \geq 0 \quad (19d)$$

$$\boldsymbol{u}(m|k\Delta t) \in \mathbb{U}, \quad \forall m \in \mathbb{N}_{[0,N-1]} \quad (19e)$$

$$\boldsymbol{e}_x(n|k\Delta t) = \boldsymbol{x}(n|k\Delta t) - \boldsymbol{x}_r(n|k\Delta t) \quad (19f)$$

$$\boldsymbol{x}(0|k\Delta t) = \tilde{\boldsymbol{x}}(k\Delta t), \quad \forall n \in \mathbb{N}_{[0,N]} \quad (19g)$$

where

$$J_N(\boldsymbol{e}_x,\boldsymbol{u}) = \sum_{n=0}^{N-1} \|\boldsymbol{e}_x(n|k\Delta t)\|_Q + \|\boldsymbol{u}(n|k\Delta t)\|_R + V(\boldsymbol{e}_x(N|k\Delta t)). \quad (20)$$

with $V(\boldsymbol{e}_x(N|k\Delta t))$ being a positive definite function of the state error. The solution to the CMPC is the optimal solution of the finite horizon optimal control problem (FHOC) in (19). Such solution is an optimal control trajectory $\boldsymbol{u}^*_k = [\boldsymbol{u}^*(0|k\Delta t),\ldots\boldsymbol{u}^*((N-1)|k\Delta t)]$ and a corresponding optimal state trajectory $\boldsymbol{x}^* = [\boldsymbol{x}^*(0|k\Delta t),\ldots\boldsymbol{x}^*_k((N-1)|k\Delta t)]$ so that the cost function $J_N(\boldsymbol{e},\boldsymbol{u})$ is minimised along the N-steps receding horizon of the FHOC problem. In (19), the constraint (19b) forces the state to evolve according to the nominal discrete dynamics of each inspector, (19c) and (19d) are the SD-HOCBF constraint on the velocity and position respectively, (19e) is the control constraint and (19g) sets the initial state of the state trajectory $\boldsymbol{x}^*$ to be equal to the measured state $\tilde{\boldsymbol{x}}(k\Delta t)$. Once the solution $\boldsymbol{u}^*_k$ to (19) is found, the feedback control $K(\tilde{\boldsymbol{x}},k\Delta t) = \boldsymbol{u}^*(0|k\Delta t)$ is applied in a ZOH fashion on (2) during the interval $[k\Delta t,(k+1)\Delta t]$. Given that at time instant $k\Delta t$ we have $\tilde{\boldsymbol{x}} \in \mathcal{C}_{S,x}(k\Delta t)$, it guaranteed that $\tilde{\boldsymbol{x}}(t) \in \mathcal{C}_{S,x}(t)\,\forall t \in [k\Delta t,(k+1)\Delta t]$ under the feedback law $K(\tilde{\boldsymbol{x}},k\Delta t)$ by direct application of Lemma 2.

Given the maximum control input ϵ_u and a sampling interval Δt, the recursive feasibility of the given CMPC scheme can be assessed numerically by applying the feasibility checking algorithm by [12]. Namely, we can verify the compatibility of constraints (19c)-(19e) by solving the following convex quadratic program

$$\min_{\boldsymbol{q},\boldsymbol{u}} -\boldsymbol{q}^T\boldsymbol{q} \quad (21a)$$

$$\boldsymbol{I}_2\boldsymbol{q} \geq \boldsymbol{0}\ , \|\boldsymbol{u}\|^2 \leq \epsilon_u^2 \tag{21b}$$

$$\zeta_{\delta\boldsymbol{r}}(\boldsymbol{x}, t, \boldsymbol{u}) \geq \bar{L}^l_{\delta\boldsymbol{r}}\Delta t + \bar{c}^l_{\delta\boldsymbol{r}}\epsilon_d + t_1 \tag{21c}$$

$$\zeta_{\delta\boldsymbol{v}}(\boldsymbol{x}, t, \boldsymbol{u}) \geq \bar{L}^l_{\delta\boldsymbol{v}}\Delta t + \bar{c}^l_{\delta\boldsymbol{v}}\epsilon_d + t_2, \tag{21d}$$

over a dense discretization of $\boldsymbol{x}$ and t, yielding a seven-dimensional parameter space for problem (21). Note that $\boldsymbol{q} = [q_1, q_2]^T$ is only a slack variable that defines how robustly (21c)-(21d) can be satisfied. We propose to lower the dimensionality of the problem by offering a more conservative solution, based on the fact that $\boldsymbol{f}_v(\tilde{\boldsymbol{x}}, t) \leq \epsilon_f \forall(\tilde{\boldsymbol{x}}, t) \in \tilde{\mathbb{X}} \times \mathbb{I}$. In particular, we note that both $\zeta_{\delta\boldsymbol{r}}$ and $\zeta_{\delta\boldsymbol{v}}$ are directly functions of the $\boldsymbol{x}$ and $\boldsymbol{t}$ through $\boldsymbol{f}_v(\boldsymbol{x}, t)$. By upper bounding $\boldsymbol{f}_v(\boldsymbol{x}, t)$ with ϵ_f, we can assess the feasibility of the CMPC scheme by solving a new convex quadratic program parameterised over $\boldsymbol{e}_{\delta\boldsymbol{v}}$ and $\boldsymbol{e}_{\delta\boldsymbol{r}}$ instead of $\boldsymbol{x}$ and t [12, Eq. 11]

$$\min_{\boldsymbol{q},\boldsymbol{u}} -\boldsymbol{q}^T\boldsymbol{q} \tag{22a}$$

$$\boldsymbol{I}_2\boldsymbol{q} \geq \boldsymbol{0}\ , \|\boldsymbol{u}\|^2 \leq \epsilon_u^2 \tag{22b}$$

$$\begin{aligned} &-2\|\tilde{\boldsymbol{e}}_{\delta\boldsymbol{v}}\|^2 - 2\|\tilde{\boldsymbol{e}}_{\delta\boldsymbol{r}}\|\epsilon_f + 2\tilde{\boldsymbol{e}}^T_{\delta\boldsymbol{r}}\boldsymbol{u} - \\ &2(p_{\delta\boldsymbol{r}0} + p_{\delta\boldsymbol{r}1})(\tilde{\boldsymbol{e}}^T_{\delta\boldsymbol{r}}\tilde{\boldsymbol{e}}_{\delta\boldsymbol{v}}) + p_{\delta\boldsymbol{r}0}p_{\delta\boldsymbol{r}1}(\epsilon^2_{\delta\boldsymbol{r}} - \|\tilde{\boldsymbol{e}}_{\delta\boldsymbol{r}}\|^2) \\ &\geq \bar{L}^l_{\delta\boldsymbol{r}}\Delta t + \bar{c}^l_{\delta\boldsymbol{r}}\epsilon_d + t_1 \end{aligned} \tag{22c}$$

$$\begin{aligned} &-2\|\tilde{\boldsymbol{e}}_{\delta\boldsymbol{v}}\|\epsilon_f - 2\tilde{\boldsymbol{e}}^T_{\delta\boldsymbol{v}}\boldsymbol{u} + p_{\delta\boldsymbol{v}0}(\epsilon^2_{\delta\boldsymbol{v}} - \|\tilde{\boldsymbol{e}}_{\delta\boldsymbol{v}}\|^2) \\ &\geq \bar{L}^l_{\delta\boldsymbol{v}}\Delta t + \bar{c}^l_{\delta\boldsymbol{v}}\epsilon_d + t_2. \end{aligned} \tag{22d}$$

where constraints (22c) and (22d) are a modified version of the SD-HOCBF constraints in (19c),(19d). More specifically, the terms $-\tilde{\boldsymbol{e}}^T_{\delta\boldsymbol{r}}\boldsymbol{f}_v$ and $-\tilde{\boldsymbol{e}}^T_{\delta\boldsymbol{v}}\boldsymbol{f}_v$ as $-\|\tilde{\boldsymbol{e}}_{\delta\boldsymbol{r}}\|\epsilon_f$ and $-\|\tilde{\boldsymbol{e}}_{\delta\boldsymbol{v}}\|\epsilon_f$ are lower bounded, such that $\zeta_{\delta\boldsymbol{v}}$ and $\zeta_{\delta\boldsymbol{r}}$ are maximally negatively decreased by the dynamic acceleration $\boldsymbol{f}_{\delta\boldsymbol{v}}$. With such formulation, the satisfaction of (22) only depends on three main parameters: the position tracking error magnitude $\|\boldsymbol{e}_{\delta\boldsymbol{r}}\|$, the velocity tracking error magnitude $\|\boldsymbol{e}_{\delta\boldsymbol{v}}\|$ and the angle $\alpha = \arccos\left(\frac{\boldsymbol{e}_{\delta\boldsymbol{v}}\cdot\boldsymbol{e}_{\delta\boldsymbol{v}}}{\|\boldsymbol{e}_{\delta\boldsymbol{r}}\|\|\boldsymbol{e}_{\delta\boldsymbol{v}}\|}\right)$. We can then iteratively check for the infeasibility of (22) over a dense discretization of $[0, \epsilon_{\delta\boldsymbol{r}}] \times [0, \epsilon_{\delta\boldsymbol{v}}] \times [0, \pi]$, which is only a three-dimensional grid instead of seven-dimensional one. We remark the fact that both ϵ_u and Δt are typically constrained by the specific hardware at hand, but they could also be treated as design parameters by increasing the dimensionality of the feasibility test by two.

VI. RESULTS

In this section, we simulate the relative dynamics of three inspectors as they orbit relative to the ISS to accomplish the inspection mission (Fig. 3). The simulation is fully developed in *Python* and we use the optimization library *CasADi* to solve the nonlinear optimal control scheme (19) by means of the Interior Point Method algorithm. The simulations were performed on a 2.3 GHz Dual-Core Intel Core i5 CPU with 8GB of RAM. The simulation environment in which the inspector and the ISS are propagated takes into account zonal harmonic terms up to order six and exponential model atmospheric drag. The inspectors are assumed to be 6U-CubeSats with a total mass of 10 kg and an omnidirectional propulsion system delivering a maximum thrust of 200 mN (and hence 0.02 ms^{-2} in acceleration). These same specifications are applied in [6] for a similar mission and are considered to give a realistic operative scenario for the inspection mission. The ISS has a mass of 419400 kg and no actuation capabilities. The reference ISS orbit parameters are obtained from New Horizon Database on the date 2023-Feb-04 00h:00m:00s. We assume the inspectors to be already deployed on three separate PRO trajectories (Fig. 3) according to the parameters in Tab II, with a non-zero initial tracking error contained inside the initial safe set. Namely, the initial condition at time $t_0 = 0$ for each inspector are $\tilde{\boldsymbol{x}}_1(0) = [55.70, 1.08, 2.43, 1.73 \times 10^{-2}, -9.23 \times 10^{-2}, 8.00 \times 10^{-3}]$, $\tilde{\boldsymbol{x}}_2(0) = [67.72, 3.27, 3.88, -2.5 \times 10^{-3}, -1.36 \times 10^{-1}, 7.01 \times 10^{-2}]$ and $\tilde{\boldsymbol{x}}_3(0) = [82.63, 0.65, 4.21, -1.39 \times 10^{-2}, -4.85 \times 10^{-2}, 1.61 \times 10^{-1}]$. Moreover, we have $\mathbb{U} = \{\boldsymbol{u} \in \mathbb{R}^3 : \|\boldsymbol{u}\| \leq 0.02\}$ and $\mathbb{I} = [0, T]$ with T being the period of the assigned PRO (which is approximately 93 min). The set $\tilde{\mathbb{X}}$ is chosen for each inspector $i = 1, 2, 3$ as $\tilde{\mathbb{X}}_i = \{\tilde{\boldsymbol{x}} \in \mathbb{R}^6 : \|\delta\boldsymbol{r}\| \leq k_1\bar{r}_{r,i} \wedge \|\delta\boldsymbol{v}\| \leq k_2\bar{v}_{r,i}\}$ with $k_1, k_2 \in \mathbb{R}_{\geq 0}$ being two sufficiently large constants that realistically capture the real workspace of each inspector. For this simulation, we applied $k_1 = k_2 = 1.4$. Table. II summarises all the CMPC parameters applied for the mission. We considered $Q = diag([50\boldsymbol{I}_{3\times3}, 59.17\boldsymbol{I}_{3\times3}])$, $R = 50\boldsymbol{I}_{3\times3}$ and $V(\boldsymbol{e}_x) = 2\boldsymbol{e}_x^T P \boldsymbol{e}_x$ where P is obtained by solving the discrete algebraic Riccati equation $P = A^T PA - (A^T PB)(R + B^T PB)^{-1}(B^T PA) + Q$. The matrix A is obtained as the dynamic matrix of the discrete-time CW model according to [6, Eq. 41] (which is the discrete-time linearised version of the nonlinear

Parameter	Inspector 1	Inspector 2	Inspector 3	units
$\bar{L}^l_{\delta\boldsymbol{v}}$	1.315×10^{-3}	1.410×10^{-3}	1.568×10^{-3}	–
$\bar{L}^l_{\delta\boldsymbol{r}}$	5.036×10^{-2}	5.414×10^{-2}	6.038×10^{-2}	–
$\bar{c}^l_{\delta\boldsymbol{v}}$	2.702×10^{-1}	2.703×10^{-1}	2.704×10^{-1}	–
$\bar{c}^l_{\delta\boldsymbol{r}}$	1.405×10^{1}	1.406×10^{1}	1.407×10^{1}	–
Δt	1.000×10^{-1}	1.000×10^{-1}	1.000×10^{-1}	s
ϵ_f	8.872×10^{-4}	1.254×10^{-3}	1.860×10^{-3}	ms^{-2}
$\epsilon_{\delta\boldsymbol{r}}$	7.000×10^{0}	7.000×10^{0}	7.000×10^{0}	m
$\epsilon_{\delta\boldsymbol{v}}$	1.330×10^{-1}	1.330×10^{-1}	1.330×10^{-1}	ms^{-1}
$\bar{\epsilon}_{\delta\boldsymbol{r}}$	7.025×10^{0}	7.029×10^{0}	7.037×10^{0}	m
$\bar{\epsilon}_{\delta\boldsymbol{v}}$	1.351×10^{-1}	1.351×10^{-1}	1.352×10^{-1}	ms^{-1}
ϵ_d	1.577×10^{-6}	2.205×10^{-6}	3.243×10^{-6}	ms^{-2}
$p_{\delta r_0}$	2.000×10^{-2}	2.000×10^{-2}	2.000×10^{-2}	–
$p_{\delta r_1}$	5.000×10^{-2}	5.000×10^{-2}	5.000×10^{-2}	–
$p_{\delta v_0}$	5.000×10^{-2}	5.000×10^{-2}	5.000×10^{-2}	–
$\bar{a}$	2.089×10^{-2}	2.126×10^{-2}	2.186×10^{-2}	ms^{-2}
$\bar{a}_r$	$1.266\ \times10^{-4}$	1.789×10^{-4}	$2.653\ \times10^{-4}$	ms^{-2}
$\bar{v}_r$	1.125×10^{-1}	1.590×10^{-1}	2.358×10^{-1}	ms^{-1}
ϵ_u	2.000×10^{-2}	2.000×10^{-2}	2.000×10^{-2}	ms^{-2}
β	6.023×10^{-4}	8.236×10^{-4}	1.189×10^{-3}	ms^{-3}
$\bar{r}_r$	1.000×10^{1}	1.414×10^{1}	2.096×10^{1}	m
ρ_r	5.000×10^{1}	6.400×10^{1}	7.800×10^{1}	m
ρ_s	0.000×10^{0}	0.000×10^{0}	0.000×10^{0}	m
ρ_w	0.000×10^{0}	6.000×10^{1}	1.400×10^{1}	m
α_w	0.000×10^{0}	0.000×10^{0}	0.000×10^{0}	rad
α_r	1.571×10^{0}	1.571×10^{0}	1.571×10^{0}	rad

TABLE II: Parameters value for the three different agents.

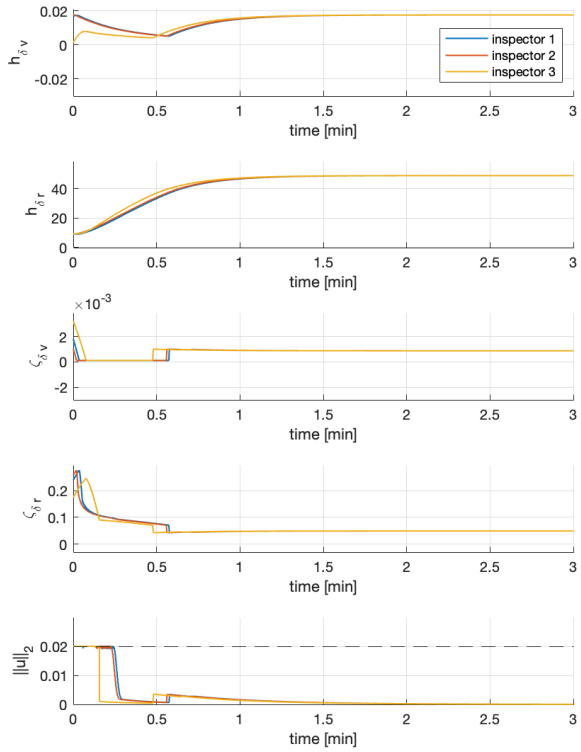

Fig. 2: Time evolution of the barrier functions $h_{\delta \boldsymbol{v}}$,$h_{\delta \boldsymbol{r}}$; the HOCBF nominal function $\zeta_{\delta \boldsymbol{r}}$,$\zeta_{\delta \boldsymbol{v}}$; and the control signal $\|\boldsymbol{u}\|_2$ for each inspector.

dynamics in (1)). The time step was chosen to be equal to $\Delta t = 0.1$s with $N = 25$ horizon steps. The simulation results can be visualized in Fig.2 for a time interval of 3 min.

We observe that each inspector is able to simultaneously satisfy (21c)-(21d) such that the safety specification in Problem 1 is respected. Particularly, we notice how the optimal control strategy (in the sense of the CMPC cost) results in an accelerate-coast-break profile: the velocity error is first increased to its maximum norm (to lower the position tracking error as fast as possible), then the inspector is left under minimum actuation until the velocity and position errors are finally driven to zero.

VII. Conclusions and Future Work

In this work, we developed new definitions of sampled-data HOCBF thanks to which continuous-time safety guarantees can be ensured for sampled-data high-order systems. Then, we explored the applicability of the CMPC control method with the newly introduced definition of SD-HOCBF for a realistic space mission scenario. In future work, we will further investigate possible algorithms to programmatically determine suitable parameters to be applied in the CMPC scheme and the usage of other noise models, and evaluate their impact on the CMPC performance.

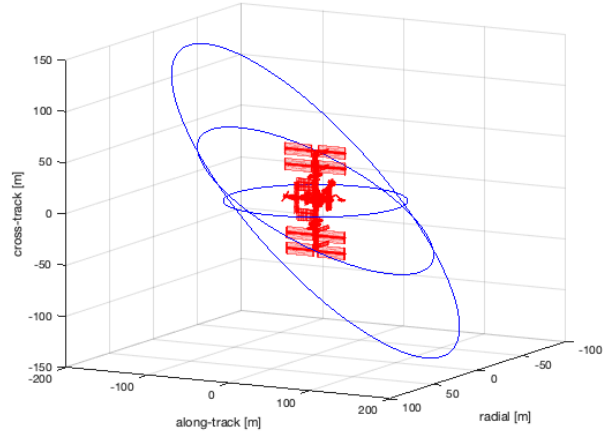

Fig. 3: Visualization of the three PRO assigned to each inspector around the ISS. The base directions are given in terms of the LVLH frame where $\hat{\boldsymbol{r}}$ is the *radial* direction, $\hat{\boldsymbol{s}}$ is the *along-track* direction and $\hat{\boldsymbol{w}}$ is the *cross-track* direction.

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